

Java Basics Interview Questions

1. What is Java?

Java is a high-level, object-oriented, platform-independent programming language developed by Sun Microsystems (now Oracle). It is used to build web, desktop, mobile, and enterprise applications.

2. Why is Java platform independent?

Java is platform independent because it is compiled into bytecode, which runs on any operating system that has a Java Virtual Machine (JVM).

3. Difference between JDK, JRE, and JVM.

JDK: Used to develop Java applications.

JRE: Provides the environment to run Java applications.

JVM: Executes Java bytecode.

4. What is bytecode?

Bytecode is the intermediate code generated by the Java compiler (.class file) that is executed by the JVM.

5. What is JVM?

JVM (Java Virtual Machine) is responsible for loading, verifying, and executing Java bytecode while managing memory and garbage collection.

6. Explain WORA.

WORA stands for 'Write Once, Run Anywhere'. Java programs can run on any platform with a compatible JVM.

7. What are primitive data types?

Java has 8 primitive data types: byte, short, int, long, float, double, char, and boolean.

8. Difference between print() and println().

print() displays output on the same line, while println() displays output and moves the cursor to the next line.

9. What is type casting?

Type casting is the process of converting one data type into another.

10. Explain implicit and explicit casting.

Implicit casting (widening) happens automatically.

Explicit casting (narrowing) is done manually using a cast operator.

11. What is a variable?

A variable is a named memory location used to store data.

12. What is an identifier?

An identifier is the name given to variables, methods, classes, packages, or interfaces.

13. Explain Java naming conventions.

Classes: PascalCase

Methods & Variables: camelCase

Constants: UPPER_CASE

Packages: lowercase.

14. What are operators in Java?

Operators are symbols used to perform operations on data, such as arithmetic, logical, relational, assignment, unary, bitwise, ternary, and shift operators.

15. What is the use of Scanner?

The Scanner class (`java.util.Scanner`) is used to read user input from the keyboard.